

2.6 - Implicit Differentiation

Goals: Compute derivatives of functions define implicitly

Before Class Videos: 2.6.1

Example 1: Explore $x^2 + y^2 = 1$. Where does it have vertical and horizontal tangents?

Big Idea: Treat $y = y(x)$ when x is independent and y is dependent.

Example 2: Find $\frac{dy}{dx}$ implicitly for $\sin(xy) + x = y^2 \cos x$.

Example 3: Compute $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = 4$.

Example 4: Now explore $x^3 + y^3 - 9xy = 0$. Find all vertical and horizontal tangents.

Logarithmic Differentiation: Recall we previously learned that $\frac{d}{dx} \ln(x) = \frac{1}{x}$.

Example 5: $y = x^x$.

Example 6:

$$y = \frac{(x+2)^3(x-1)^{1/3}}{(x+7)^2}.$$

Example 7:

$$y = x^{\cos x} \cdot (\cos x)^x.$$